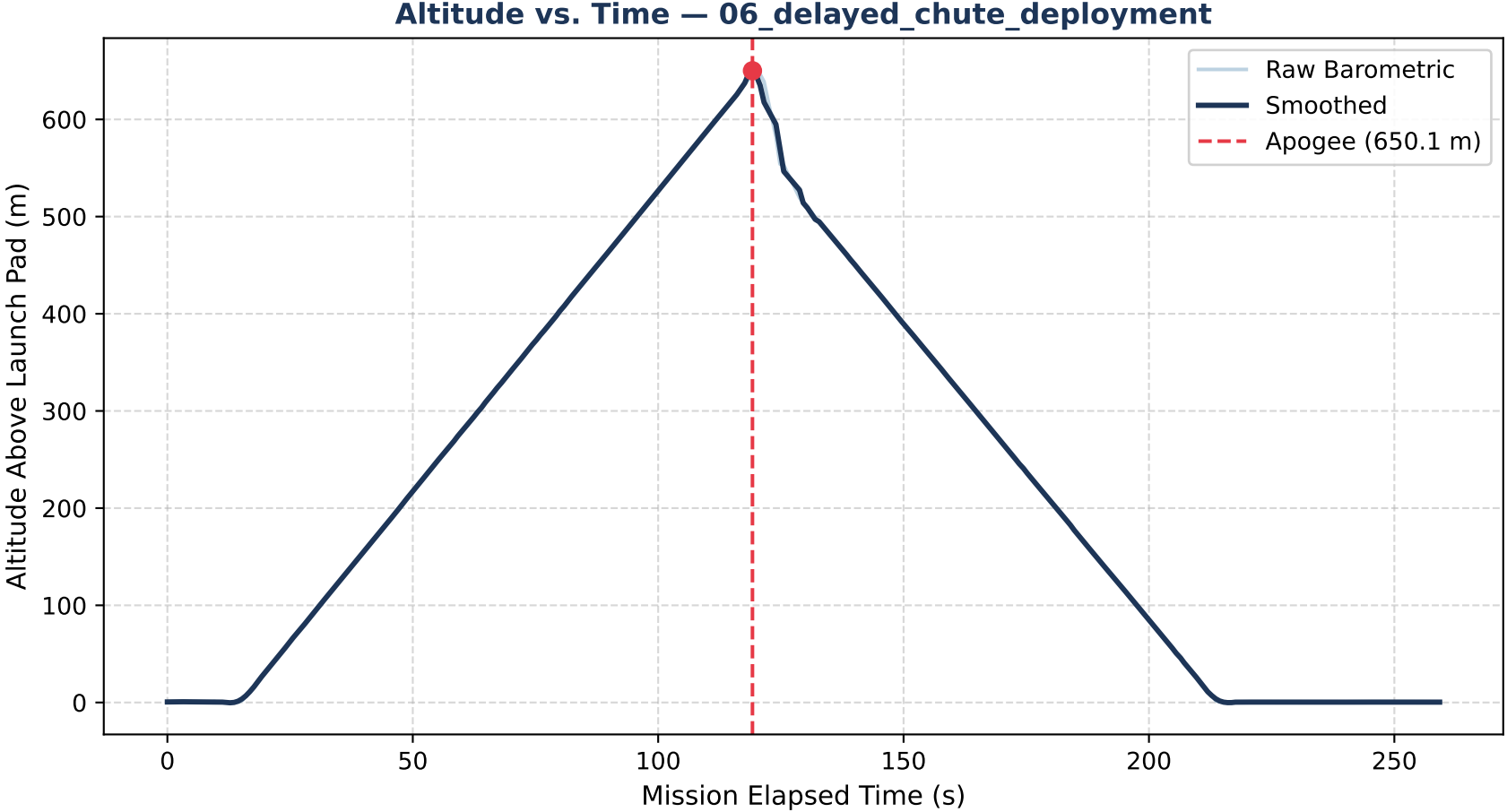
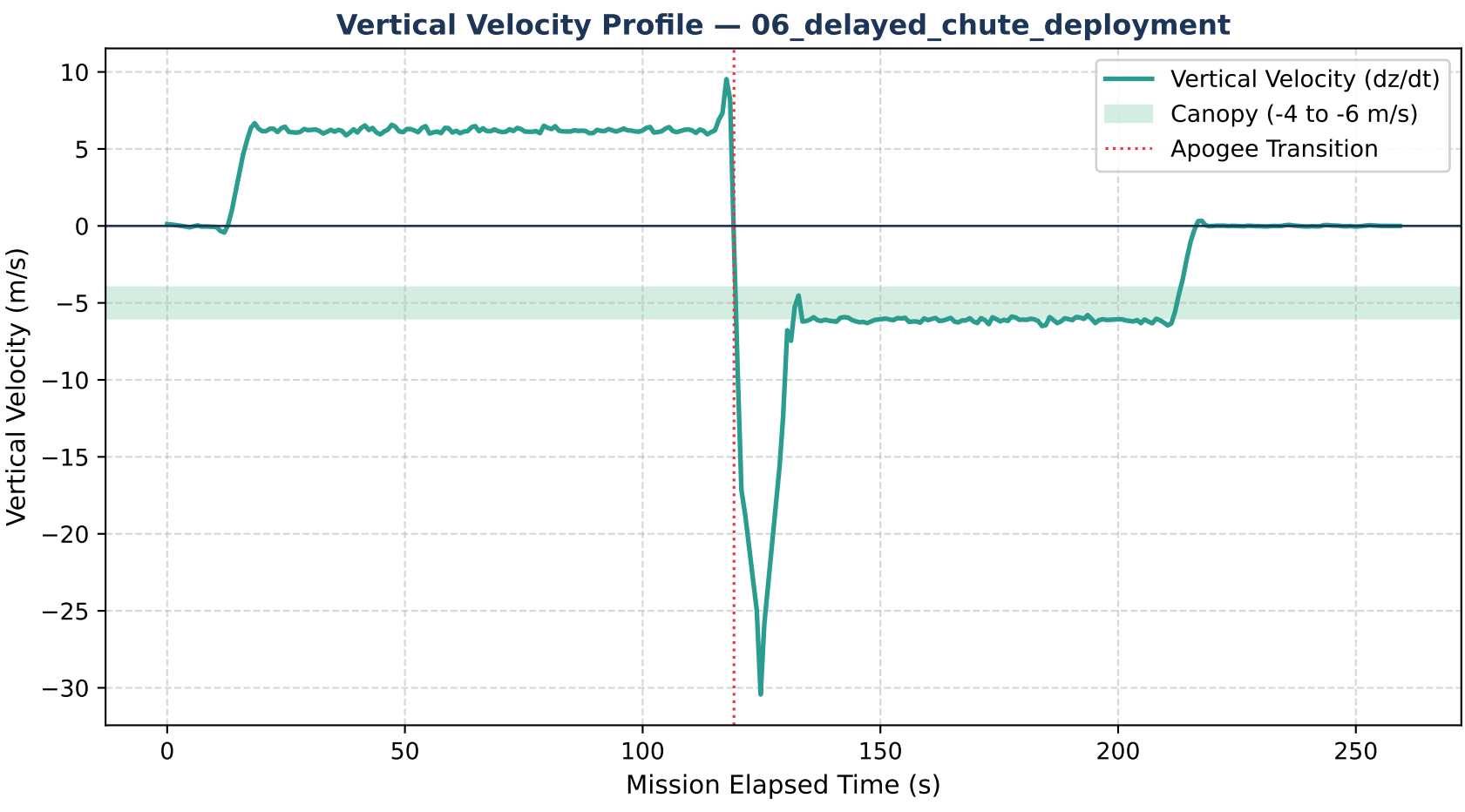
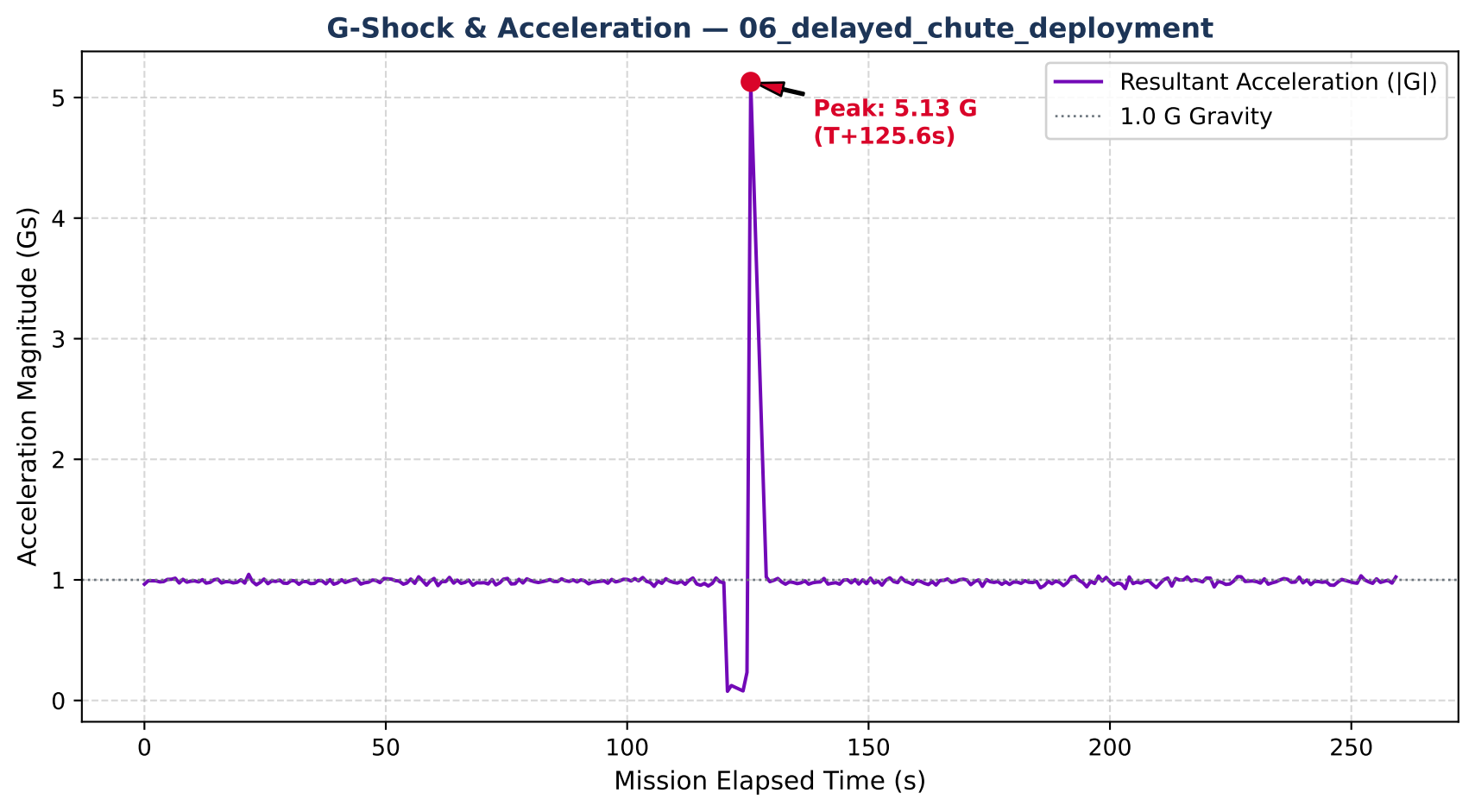
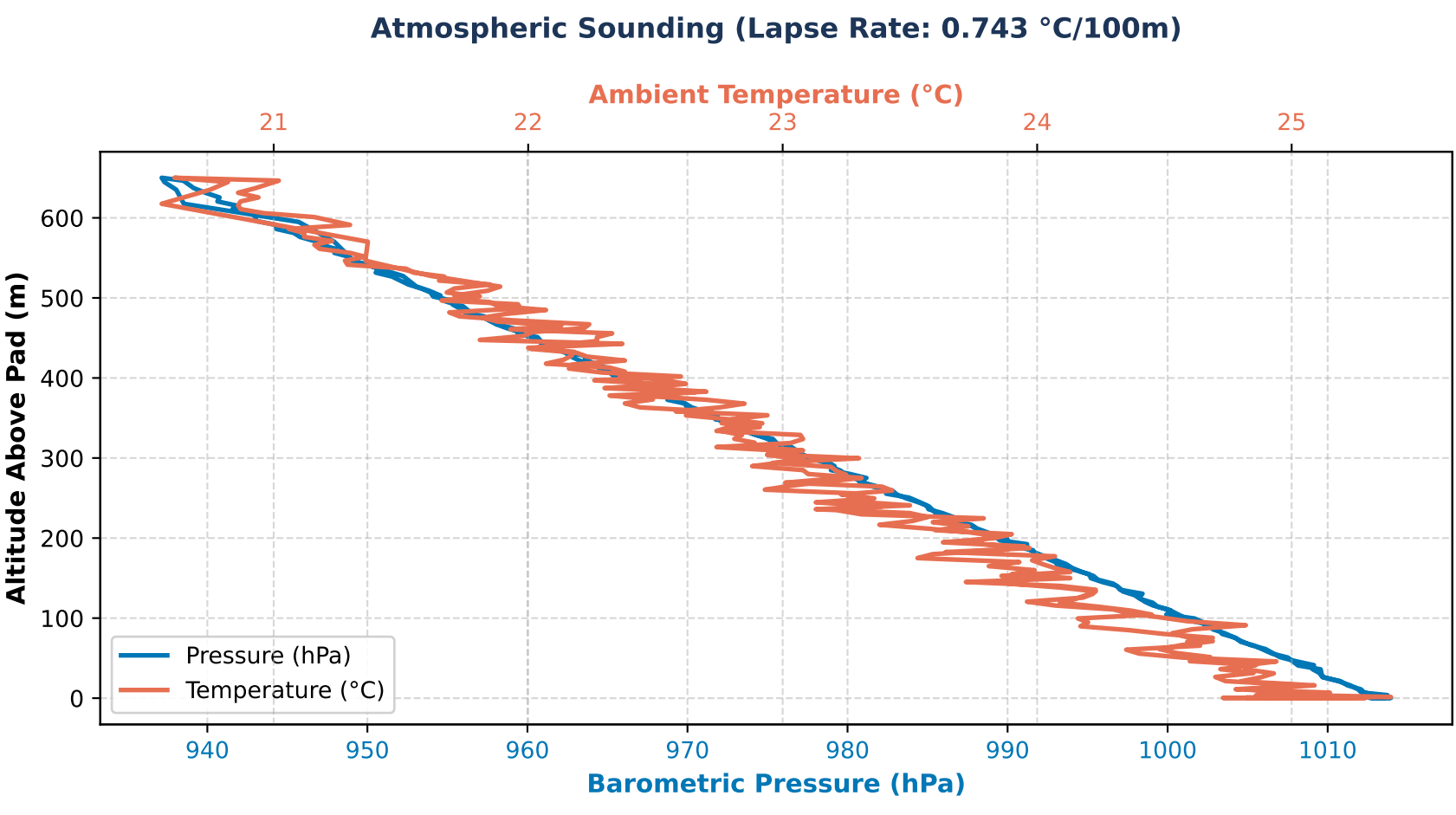
Cognitive CanSat - Mission Summary: 06_delayed_chute_deployment

Flight Dynamics Parameter	Calibrated Value
Mission Name	06_delayed_chute_deployment
Apogee (m)	650.09
Time to Apogee (s)	119.21
Max Ascent Velocity (m/s)	9.53
Max Descent Velocity (m/s)	-30.43
Max G-Shock (Gs)	5.13
Touchdown Shock (Gs)	1.03
Mission Duration (s)	259.22
Min Temperature (°C)	20.56
Max Temperature (°C)	25.39
Min Pressure (hPa)	937.15
Max Pressure (hPa)	1013.95
Lapse Rate (°C/100m)	0.743
Telemetry Frames	320
Landing Lat	22.574934
Landing Lon	88.368117


Altitude vs. Time — 06_delayed_chute_deployment
Raw Barometric
Smoothed
Apogee (650.1 m)
Altitude Above Launch Pad (m)
0
100
200
300
400
500
600
0
50
100
150
200
250
Mission Elapsed Time (s)


Vertical Velocity Profile — 06_delayed_chute_deployment
Vertical Velocity (dz/dt)
Canopy (-4 to -6 m/s)
Apogee Transition
Vertical Velocity (m/s)
Mission Elapsed Time (s)
10
5
0
−5
−10
−15
−20
−25
−30
0
50
100
150
200
250


G-Shock & Acceleration — 06_delayed_chute_deployment
Resultant Acceleration (|G|)
1.0 G Gravity
Peak: 5.13 G
(T+125.6s)
Acceleration Magnitude (Gs)
Mission Elapsed Time (s)
0
1
2
3
4
5
0
50
100
150
200
250


Atmospheric Sounding (Lapse Rate: 0.743 °C/100m)
Ambient Temperature (°C)
21
22
23
24
25
Altitude Above Pad (m)
600
500
400
300
200
100
0
Pressure (hPa)
Temperature (°C)
940
950
960
970
980
990
1000
1010
Barometric Pressure (hPa)

Cognitive CanSat Mission Dynamics: 06_delayed_chute_deployment

